

Federated feature selection with false discovery rate control

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Abstract

Selecting a set of universally relevant features associated with a given response variable across multiple distributed data sites is an important problem in numerous scientific fields. However, performing this federated feature selection task becomes challenging when individual-level data cannot be shared due to privacy concerns. The problem is further complicated by potential heterogeneity in both feature distributions and model parameters across sites. In this paper, we propose Fed-false discovery rate (FDR), a federated feature selection framework that simultaneously identifies important features while controlling the FDR. To ensure privacy preservation and reduce communication costs, the Fed-FDR shares only lower-dimensional coefficient estimates instead of transmitting summary statistics for all features, with the dimensionality shown to be of the same order as the number of relevant features. The coordinating centre then leverages these lower-dimensional coefficient estimates to construct a generalized mirror statistic to identify the important features. The Fed-FDR is robust to the heterogeneity of feature distribution and model parameters, easy to implement, and computationally efficient. We further demonstrate that Fed-FDR effectively controls the FDR while achieving strong statistical power in our simulation studies. The results of the empirical study also demonstrate that the method is both valid and implementation-ready.

Keywords: communication efficiency, computational efficiency, GLM model, generalized mirror statistics, privacy protection

1. Introduction

Recent technological advancements have empowered researchers to collect a multitude of explanatory features targeting a specific response variable, with the aim of avoiding losing valuable information. At the same time, many null features are included, which not only exacerbate the complexity of the learning algorithm, but also degrade model accuracy and interpretability. A widely held belief is that the response variable primarily relies on a limited subset of the collected features. Identifying these key relevant features is therefore essential, as it can significantly enhance computational efficiency, improve estimation accuracy, and strengthen the interpretability of scientific discoveries (Kumar & Minz, 2014; Miao & Niu, 2016). In many real-world applications, however, data are not centrally stored but instead distributed across multiple sites. This work is motivated by an empirical analysis in Section 6, in which we apply the newly proposed procedure to a synthetic dataset derived from the summary statistics from electronic health records (EHR) data in existing studies (Thaweethai et al., 2023; Wu et al., 2024), to identify the important

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features associated with SARS-CoV-2 infection in children during the Omicron period. The final simulated data for analysis consists of 3,990 individuals collected from 34 clinical sites, with each site contributing approximately 130 patients. Research networks offer researchers the opportunity to integrate data from multiple sites, enhancing statistical power and broadening the applicability of scientific findings. However, the distributed data also introduce additional significant challenges for federated feature selection.

There are several key challenges in selecting the relevant features in multi-site studies. The first one lies in the difficulty of sharing patient-level data, which is often restricted by privacy concerns and varying policy regulations in biomedical research (Behlen & Johnson, 1999). To address this challenge, researchers have proposed various distributed methods, using the idea of ‘divide and conquer’ (Battey et al., 2018; Lee et al., 2017). Nevertheless, these methods primarily focus on estimating averaged high-dimensional coefficients rather than addressing the feature selection aspect. The second significant challenge involves the presence of population heterogeneity. One form of heterogeneity, known as covariate shift, occurs when the distribution of covariates varies across different sites. Another form, model shift, arises when the conditional distribution of the response given the covariates differs between sites. Such variations are particularly common in clinical settings, where factors such as geographical differences in disease prevalence, patient demographics, and regional practice patterns contribute to site-specific differences. Ignoring these disparities can lead to biased estimation. Cai et al. (2022) have introduced a high-dimensional integrative regression (SHIR) approach that can handle the aforementioned challenges through a summary statistics-based integration procedure, while also achieving consistent variable selection. While this method exhibits mathematical elegance, it suffers from the third challenge of high dimensionality. It requires the transmission of gradients and Hessian matrices from each site to the coordinating centre, which could be communication expensive, especially when dealing with extremely high-dimensional settings. In such cases, even transferring a high-dimensional gradient can be time-consuming, logistically complex, or even practically infeasible. Similar issues arise in the methods proposed by Liu et al. (2021) and Yan et al. (2023), both of which fail to achieve a sufficiently low communication cost for real-world applications, leaving the challenge of managing communication budgets unresolved. Alongside these challenges, a desirable feature selection procedure is expected to have the capability to control the false discovery rate (FDR) (Benjamini & Hochberg, 1995). The FDR is defined as $\text{FDR} = \mathbb{E}(\text{FDP})$, where $\text{FDP} = |\hat{S} \cap S_0| / (|\hat{S}| \vee 1)$ with S_0 denotes the index set of null features, $\hat{S}$ represents the set of selected features, $a \vee b = \max\{a, b\}$, and FDP refers to the ‘false discovery proportion’. The expectation is taken over the randomness in both the data and the selection procedure.

Commonly used FDR control procedures include Benjamin–Hochberg (BHq) procedure (Benjamini & Hochberg, 1995) and Knock-off procedure (Barber & Candès, 2015). The BHq method relies on p -values and ensures exact FDR control when all the p -values are independent. Benjamini and Yekutieli (2001) then extended the BHq procedure to handle dependent p -values. However, constructing p -values becomes challenging in high-dimensional settings. To address this, Javanmard and Javadi (2019) and Ma et al. (2021) explored the application of BHq to high-dimensional linear and logistic regression models, respectively, with p -values derived using the desparsified Lasso method (Van de Geer et al., 2014). However, the desparsified Lasso typically requires regularized node-wide regressions, which can be computationally intensive, and the resulting estimators are no longer sparse. Thus, the BHq method is not well suited for the purpose of this paper. Barber and Candès (2015) first introduced the fixed-design knock-off. Sarkar and Tang (2022) further integrated the seminal ideas of knock-off and BHq to control the FDR without relying on any specific correlation structure of the explanatory variables. Both methods are tailored for low-dimensional Gaussian linear models. Candès et al. (2018) proposed the model-X knock-off, which extends the utility of knock-off to high-dimensional problems. Huang and Janson (2020) further generalized the model-X knock-off method by introducing conditioning, enabling the inclusion of features following an exponential family distribution with unknown parameters. Cao et al. (2024) considered split knock-offs for controlling FDR under transformational sparsity. Despite their strengths, knock-off-based methods often require precise or partial knowledge of the joint distribution of all features, which limits their practical applicability.

In this paper, we propose Fed-FDR, a novel framework for federated feature selection with FDR control. Instead of sharing summary statistics for all features, such as gradients or Hessian matrices (Cai et al., 2022), Fed-FDR allows the coordinating centre to construct generalized mirror statistics using only lower-dimensional coefficient vector estimates, whose dimensionality is proven to be on the same order as the number of relevant features. The generalized mirror statistics, defined in (2), include the mirror statistics proposed by Dai et al. (2023a) as a special case. The lower-dimensional coefficient vector estimates are derived through a two-step process. Firstly, a generalized linear model (GLM) Lasso estimator is applied to identify the subset of relevant features. Secondly, a refined desparsified Lasso (Xing et al., 2023) is utilized in the context of lower-dimensional GLMs that were selected in the first step. This process ensures a high probability of direct invertibility of the lower-dimensional Hessian matrix identified in the first step, eliminating the need for regularized node-wise regressions as employed in Dai et al. (2023b). Consequently, the proposed method significantly reduces both computation and communication costs. Moreover, Fed-FDR offers several advantages: it is robust to unknown feature distributions, easy to implement, and computationally efficient, with extremely low communication costs, making it highly practical for real-world applications. We summarize our contributions as follows.

- Firstly, Fed-FDR is communication-efficient and privacy-preserving, as it only requires transferring a lower-dimensional subset of the coefficient estimates in one round of communication. Moreover, the dimension of the subset coefficient estimates is demonstrated to be of the same order as the number of relevant features, which minimize both the communication cost and the potential of data privacy leakage.
- Secondly, our method efficiently handles high-dimensional data, where the response variable can be non-Gaussian, and avoids the use of regularized node-wise Lasso, significantly reducing computational complexity.
- Thirdly, Fed-FDR accounts for heterogeneity in both the feature populations (e.g. differences in population or systematic variations) and model parameters across multi-sites. Its robustness in addressing these two types of heterogeneity makes it highly suitable for analysing real-world clinical data.

To the best of our knowledge, this paper is the first to offer an approach to address the challenge task of federated feature selection with FDR control, while simultaneously minimizing the communication cost, with costs scaling is the same order as the number of relevant features, and protect the data privacy by sharing only coefficient estimates with the coordinating centre.

The rest of the paper is organized as follows. In Section 2, we formulate the problem of interest within the framework of GLMs and introduce the generalized mirror statistics for the coordinating centre to perform federated feature selection with FDR control. In Section 3, we develop a refined desparsified Lasso estimation procedure for each site to obtain a reliable estimator and reduce computation and communication costs. The asymptotic properties of the proposed framework, along with its extension to support heterogeneity and address model misspecification, are examined in Section 4. We investigate the numerical performance via extensive simulations in Section 5. In Section 6, we apply the proposed method to the simulated dataset to identify important features associated with acute sequelae of SARS-CoV-2 infection (PASC) in children during the Omicron period, considering 3,990 patients from 34 clinical sites. We conclude the paper with discussions of potential future research in Section 7. Proofs and more simulation results are deferred to [online supplementary materials](#).

2. Methodology

BHq-based procedures and knock-off-based procedures are perhaps the most popular FDR control procedures in the literature. Recently, Dai et al. (2023a) proposed an innovative approach to controlling the FDR in feature selection. We shall propose generalized mirror statistics defined in (2) that include the mirror statistics proposed by Dai et al. (2023a) as a special case. Before we pursue further, let us introduce some preliminaries about mirror statistics and desparsified Lasso.

2.1 Preliminaries: mirror statistics and desparsified Lasso

The FDR control procedure proposed by Dai et al. (2023a) involves constructing a mirror statistic M_j for each feature $j \in 1, \dots, p$, and M_j possesses the following two key properties:

- (i) A feature is more likely to be considered relevant if its mirror statistic is larger.
- (ii) The distribution of the mirror statistic under the null is (asymptotically) symmetric about zero.

Property (i) indicates that the importance of each feature can be ranked based on the mirror statistic, allowing us to select features with mirror statistics exceeding a threshold t . Property (ii) implies that the number of false positives can be conservatively estimated using the symmetry property, i.e. $\#\{j: j \in S_0, M_j > t\} \approx \#\{j: j \in S_0, M_j < -t\} \leq \#\{j: M_j < -t\}$.

The FDP(t) of the selection set $\hat{S}_t = \{j: M_j > t\}$, along with its conservative estimate, denoted as $\hat{\text{FDP}}(t)$, are given by

$$\text{FDP}(t) = \frac{\#\{j: M_j > t, j \in S_0\}}{\#\{j: M_j > t\} \vee 1}, \quad \hat{\text{FDP}}(t) = \frac{\#\{j: M_j < -t\}}{\#\{j: M_j > t\} \vee 1}.$$

For a given FDR control level $\alpha \in (0, 1)$, the data-driven cut-off t_α is chosen as

$$t_\alpha = \min \{t > 0: \hat{\text{FDP}}(t) \leq \alpha\}.$$

The final selection is then given by $\hat{S} = \{j: M_j > t_\alpha\}$. This method is robust to the unknown distribution of features, easy to implement, and computationally efficient. Dai et al. (2023b) further extended the use of mirror statistics to the GLM. However, their procedure employs the desparsified Lasso (Van de Geer et al., 2014), which has the general form $\hat{\beta}^{(d)} = \hat{\beta}^{(\lambda)} - \hat{\Theta} P_n \dot{L}_{\hat{\beta}^{(\lambda)}}$, where $\hat{\beta}^{(\lambda)}$ is the Lasso estimator, and $P_n \dot{L}_{\hat{\beta}^{(\lambda)}}$ represents the empirical average of the gradient (score) of the loss function L evaluated at $\hat{\beta}^{(\lambda)}$ over the n data points, and $\hat{\Theta}$ is typically an estimate of the inverse of the Hessian matrix. Readers are referred to Section 3. for more details. Estimating $\hat{\Theta}$ typically requires regularized node-wise regressions (see Section S.2 of the online supplementary material for details), which can be computationally intensive. Additionally, the resulting desparsified Lasso estimator is no longer sparse. Although Dai et al. (2023b)'s proposal is powerful in feature selection with controlled FDR, it is specifically designed for single dataset where individual-level data are accessible. Consequently, they cannot be directly employed to address the challenge of federated feature selection with FDR under control.

2.2 Problem setup

Suppose that the data of interest are distributed across K sites. At the k th site, we observe n_k independent and identically distributed (i.i.d.) samples, denoted by y_{ki} and $\mathbf{x}_{ki} \in \mathbb{R}^p$, for $i = 1, \dots, n_k$. For simplicity of notation and without loss of generality, we assume $n_1 = \dots = n_K = n$. We consider the high-dimensional GLMs for each site. Specifically, for site k , given $\mathbf{x}_{ki}$, the response y_{ki} follows an exponential family distribution

$$\mathbb{P}(y_{ki} | \mathbf{x}_{ki}, \boldsymbol{\beta}^{(k*)}) = c(y_{ki}) \exp\left(y_{ki} \mathbf{x}_{ki}^T \boldsymbol{\beta}^{(k*)} - \rho\left(\mathbf{x}_{ki}^T \boldsymbol{\beta}^{(k*)}\right)\right), \quad (1)$$

where $\boldsymbol{\beta}^{(k*)} \in \mathbb{R}^p$ denotes the true parameter, $c(\cdot)$ and $\rho(\cdot)$ are some known univariate functions. In commonly used GLMs, ρ exhibits strict convexity and possesses infinite differentiability.

In this paper, we are interested in selecting a common important feature support across all sites. Instead of the more restrictive homogeneous assumption that all sites share the same parameter, i.e. $\boldsymbol{\beta}^{(1*)} = \dots = \boldsymbol{\beta}^{(K*)} := \boldsymbol{\beta}$, we adopt a more flexible assumption regarding the parameters:

A1 Assume that all sites share a common baseline parameter $\beta = \{\beta_1, \dots, \beta_p\}$, with $\beta_j \in \{-1, 0, 1\}$ denotes the negative effect, no effect, or positive effect of feature j . To accommodate model shift, we allow varying signal strength across different features and sites, i.e. $\beta_j^{(k*)} = \sigma_j^k \beta_j$ for $j = 1, \dots, p$ and $k = 1, \dots, K$, where σ_j^k is the positive signal strength that indicates the effect size of the feature.

Assumption A1 allows model shift across different sites, which implies that the conditional distribution of the response given the covariates may be different from one to another. More importantly, in such setting, all the sites share a common feature support $S_1 = \{j \in \{1, \dots, p\} | \beta_j \neq 0\}$, which is of the main interest of this paper. In addition, we also consider the existence of covariate shift across sites. By covariate shift, it means that the distribution of the covariates may be different across sites.

These two types of heterogeneity are more realistic in real-world applications compared to the setting where the data across sites are assumed to be homogeneous. For example, consider EHR data for patients with a certain disease condition, such as diabetes. In this context, baseline parameters could represent the common set of clinical measurements, such as blood glucose levels, HbA1c values, and blood pressure, that are recorded for both well-managed diabetics and poorly controlled diabetics. However, the signal strength of these clinical measurements may vary between the two groups due to differences in disease management and patient compliance. Additionally, the distributions of these clinical measurements could also differ between these two types of patients.

Let $p_1 = |S_1|$ and $p_0 = |S_0| = p - |p_1|$ be the number of nonnull and null features, respectively. Hereafter, we use the power to denote the accuracy of a feature selection procedure

$$\text{Power} = \frac{\mathbb{E}|\hat{S} \cap S_1|}{|S_1|}.$$

The primary goal of this paper is to select as many relevant features shared across all sites as possible, i.e. increase power as much as possible, while ensuring the FDR is under control. Simultaneously, our method strives to protect individual-level data and minimize communication costs by sharing as minimal summary statistics as possible.

2.3 Federated feature selection via generalized mirror statistic

To protect individual-level data, each site is only required to transfer a basic summary statistic of each feature to the central site. The central site then utilizes these summary statistics to construct generalized mirror statistics for each individual feature j , facilitating the selection of relevant features while controlling the FDR. Specifically, in the context of regression settings, each site k only transfers a set of estimates, $\hat{\beta}^{(k)}$, of the true coefficient $\beta^{(k*)}$, whose distribution is (asymptotically) symmetric about $\beta^{(k*)}$. In this section, we focus on the tasks of federated feature selection at the central site, based on the coefficient estimators $\hat{\beta}^{(k)}$ for $k = 1, \dots, K$ obtained from the local sites. The discussion on how to construct a good estimator $\hat{\beta}^{(k)}$ that is (asymptotically) symmetric around $\beta^{(k*)}$ for each site is deferred to Section 3. Since $\hat{\beta}^{(s)}$ and $\hat{\beta}^{(t)}$ are naturally independent, for each pair of sites s and t satisfying $1 \leq s < t \leq K$, the central site can thus compute the generalized mirror statistic $M_j^{(st)}$ for each feature j . The generalized mirror statistic for feature j from s th and t th sites is defined as

$$M_j^{(st)} = \text{sign}(\hat{\beta}_j^{(s)} \hat{\beta}_j^{(t)}) f(|\hat{\beta}_j^{(s)}|, |\hat{\beta}_j^{(t)}|), \quad (2)$$

where $f(x, y)$ is a nonnegative, exchangeable and monotonically increasing function defined on $\mathbb{R}^+ \times \mathbb{R}^+$. In this paper, we refer to $M_j^{(st)}$ as generalized mirror statistic, since the true coefficients of each feature, $\beta_j^{(s*)} = \sigma_j^s \beta_j$ and $\beta_j^{(t*)} = \sigma_j^t \beta_j$, may differ across sites through the site-specific signal strengths σ_j^s and σ_j^t applied to the underlying effect β_j . It is worth noting that Dai et al. (2023a)

constructed the mirror statistic from a single dataset, relying on the same model, i.e. the same signal strength across all sites, and therefore represents a special case of the generalized mirror statistic. We adopt the recommendation of Dai et al. (2023a) to employ $f(x, y) = x + y$, which corresponds to the ‘sign-max’ of $|\hat{\beta}_j^{(1)} + \hat{\beta}_j^{(2)}|$ and $|\hat{\beta}_j^{(1)} - \hat{\beta}_j^{(2)}|$, and has been demonstrated to be optimal in a simplified setting, as illustrated by Dai et al. (2023a).

By definition, the generalized mirror statistic $M_j^{(\text{st})}$ satisfies two properties (i) and (ii). Based on these two properties, it is reasonable to select the features by $\hat{S}^{(\text{st})} = \{j : M_j^{(\text{st})} > \tau_\alpha^{(\text{st})}\}$, with the threshold $\tau_\alpha^{(\text{st})}$ determined as follows:

$$\tau_\alpha^{(\text{st})} = \min \left\{ \tau > 0 : \widehat{\text{FDP}}(\tau) = \frac{\#\{j : M_j^{(\text{st})} < -\tau\}}{\#\{j : M_j^{(\text{st})} > \tau\} \vee 1} \leq \alpha \right\}. \quad (3)$$

This controls the FDR at the prespecified level α without sharing any individual-level data. The selection procedure of $\hat{S}^{(\text{st})}$ can repeat $K(K-1)/2$ times. Each time we can obtain a selected features set $\hat{S}^{(\text{st})}$ for $1 \leq s < t \leq K$. Then, for each feature j , we can estimate the inclusion rate $\hat{I}_j$ as

$$\hat{I}_j = \frac{2}{K(K-1)} \sum_{1 \leq s < t \leq K} \frac{\mathbb{I}(j \in \hat{S}^{(\text{st})})}{|\hat{S}^{(\text{st})}| \vee 1}.$$

Intuitively speaking, a relevant feature is more likely to be selected in the repeated pairwise feature selection procedure, while a null feature is less likely to be selected. Therefore, similar to Dai et al. (2023a), we can rank the inclusion rates and then choose a proper threshold to control the FDR. Specifically, we can determine the final selected features as follows: firstly, sort the estimated inclusion rate by $0 \leq \hat{I}_{(1)} \leq \hat{I}_{(2)} \leq \dots \leq \hat{I}_{(p)}$; secondly, find the largest $m \leq p$ such that $\hat{I}_{(1)} + \dots + \hat{I}_{(m)} \leq \alpha$ for the given FDR level $\alpha \in (0, 1)$; finally, select the features by $\hat{S} = \{j : \hat{I}_j > \hat{I}_{(m)}\}$. The whole procedure for federated feature selection with FDR control is summarized in Algorithm 1.

To control the FDR through the utilization of the generalized mirror statistic (2), we summarize the conditions for the coefficient estimates $\hat{\beta}_j^{(k)}$ and the generalized mirror statistics $(M_j^{(\text{st})})$ as follows:

- C1 Symmetry: The distribution of the estimators $(\hat{\beta}_j^{(k)})$ ’s are (asymptotic) symmetric about zero for $j \in S_0$ and $1 \leq k \leq K$.
- C2 Weak dependence: The generalized mirror statistics $(M_j^{(\text{st})})$ ’s are continuous random variables, and there exist constants $c > 0$ and $r \in (0, 2)$ such that

$$\text{var} \left(\sum_{j \in S_0} \mathbb{I}(M_j^{(\text{st})} > m) \right) \leq cp_0^r,$$

holds for any $m \in \mathbb{R}$ and $1 \leq s < t \leq K$.

Note that condition C1 only assumes the (asymptotic) symmetry of the null features. Condition C2 requires taking into account the correlation among the null features, regardless of the correlations between the relevant features. In Section 3, we show that for GLMs, a refined desparsified Lasso method can be employed to achieve the symmetric property of the estimator $\hat{\beta}_j^{(k)}$ and reduce the communication cost to as low as $O(Kp_1)$. The weak dependence condition holds as long as the Hessian matrix of the null features satisfies certain regularity assumptions. Then, we outline how the procedure in Algorithm 1 effectively controls the FDR, as summarized in the following proposition.

Algorithm 1 Fed-FDR (the task for the central site).

Require: $\hat{\beta}^{(k)}$ for $1 \leq k \leq K$, the given FDR level $\alpha \in (0, 1)$.
Ensure: The final support $\hat{S}$.
1: **for** $s = 1, \dots, K - 1$ **do**
2: **for** $t = s + 1, \dots, K$ **do**
3: Calculate $M_j^{(st)} = \text{sign}(\hat{\beta}_j^{(s)} \hat{\beta}_j^{(t)}) f(|\hat{\beta}_j^{(s)}|, |\hat{\beta}_j^{(t)}|)$ for $1 \leq j \leq p$.
4: Select the feature $\hat{S}^{(st)} = \{j : M_j^{(st)} > \tau_\alpha^{(st)}\}$, where the threshold $\tau_\alpha^{(st)}$ is

$$\tau_\alpha^{(st)} = \min \left\{ \tau > 0 : \widehat{\text{FDP}}(\tau) = \frac{\#\{j : M_j^{(st)} < -\tau\}}{\#\{j : M_j^{(st)} > \tau\} \vee 1} \leq \alpha \right\}.$$

5: **end for**

6: **end for**

7: Calculate the inclusion rate $\hat{I}_j = \frac{2}{K(K-1)} \sum_{1 \leq s < t \leq K} \frac{I(j \in \hat{S}^{(st)})}{|\hat{S}^{(st)}| \vee 1}$.

8: Sort the estimated inclusion rate: $0 \leq \hat{I}_{(1)} \leq \hat{I}_{(2)} \leq \dots \leq \hat{I}_{(p)}$.

9: Find the largest $m \in \{1, \dots, p\}$ such that $\hat{I}_{(1)} + \dots + \hat{I}_{(m)} \leq \alpha$.

10: Obtain the final feature support $\hat{S} = \{j : \hat{I}_j > \hat{I}_{(m)}\}$.

Proposition 1 Assume $\text{var}(M_j^{(st)})$ is uniformly bounded above and away from zero, and for any given FDR level $\alpha \in (0, 1)$, there exists a constant $t_\alpha > 0$ such that $P(\text{FDP}(t_\alpha) \leq \alpha) \rightarrow 1$ as $p \rightarrow \infty$. Then, under conditions C1 and C2, the one time selection procedure based on (3) and the federated feature selection procedure in Algorithm 1 satisfies

$$\text{FDP}(\tau_\alpha^{(st)}) \leq \alpha + o_p(1) \quad \text{and} \quad \limsup_{n, p \rightarrow \infty} \text{FDR} \leq \alpha, \quad \text{respectively.}$$

The existence of $t_\alpha > 0$ such that $P(\text{FDP}(t_\alpha) \leq \alpha) \rightarrow 1$ ensures the asymptotic feasibility of controlling the FDR based on the ranking of features by their generalized mirror statistics, which also appeared in Dai et al. (2023a). The remaining task is to obtain good estimator $\hat{\beta}_j^{(k)}$ for each site and minimize the communication cost.

3. High-dimensional GLMs for each site

3.1 Overview

Based on the distribution defined in (1), for each data point $(y_{ki}, \mathbf{x}_{ki})$ and given β , we define the GLM loss function as the negative log-likelihood up to an additive constant:

$$L_\beta(y_{ki}, \mathbf{x}_{ki}) = \rho(\mathbf{x}_{ki}^T \beta) - y_{ki} \mathbf{x}_{ki}^T \beta,$$

and its first and second derivatives with respect to β are denoted as $\dot{L}_\beta$ and $\ddot{L}_\beta$, respectively. Denote $P_n \dot{L}_\beta = n^{-1} \sum_{i=1}^n \{\rho'(\mathbf{x}_{ki}^T \beta) - y_{ki}\} \mathbf{x}_{ki}$ and $P_n \ddot{L}_\beta = n^{-1} \sum_{i=1}^n \rho''(\mathbf{x}_{ki}^T \beta) \mathbf{x}_{ki} \mathbf{x}_{ki}^T$. The Lasso estimator for site k is given by

$$\hat{\beta}^{(k, \lambda_{1k})} = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{n} \sum_{i=1}^n L_\beta(y_{ki}, \mathbf{x}_{ki}) + \lambda_{1k} \|\beta\|_1 \right\}. \quad (4)$$

In general, the Lasso estimate does not follow an asymptotical normal distribution due to the L_1 regularization. Van de Geer et al. (2014) proposed desparsified Lasso method to construct an

estimate that is asymptotically normal with mean being the true value $\beta^{(k*)}$. Specifically, the desparsified Lasso estimate is defined below

$$\hat{\beta}^{(k,d)} = \hat{\beta}^{(k,\lambda_{1k})} - \hat{\Theta}_k P_n \dot{L}_{\hat{\beta}^{(k,\lambda_{1k})}}.$$

Here, $\hat{\Theta}_k$ is an estimator of the inverse of the Hessian matrix $\Sigma_{\beta^{(k*)}}^{-1} = [\mathbb{E}(P_n \ddot{L}_{\beta^{(k*)}})]^{-1}$ evaluated at the true coefficient vector $\beta^{(k*)}$. However, in high-dimensional settings, a key assumption for accurately estimating the inverse of the Hessian matrix is the sparsity assumption on $\Sigma_{\beta^{(k*)}}^{-1}$, and such sparsity assumption is difficult to interpret and often may not hold in practice. These issues were discussed further by [Xia et al. \(2023\)](#). Moreover, constructing $\hat{\Theta}$ in high dimensions typically requires the use of regularized node-wise regressions (see [Section S.2 of the online supplementary material for details](#)), which can be computationally intensive, and the resulting desparsified Lasso estimator is no longer sparse. As a result, each site may need to transfer coefficient estimates in a p -dimensional vector to the coordinating centre, which leads to high communication costs, especially when the dimensionality is large.

3.2 A between-site swap strategy

To address the aforementioned limitations of the high-dimensional desparsified Lasso estimator obtained via node-wise Lasso regressions, we propose a strategy called the ‘between-site swap’, in which the support estimates from each site are exchanged with one another to construct a lower-dimensional estimator. Specifically, for each site $k \in \{1, \dots, K\}$, we propose to first fit a GLM lasso estimator, $\hat{\beta}^{(k,\lambda_{1k})}$, using (4), where the tuning parameter λ_{1k} is selected via fivefold cross-validation (CV). This process yields the selected support set $\hat{S}^{(k)} = \{j : \hat{\beta}_j^{(k,\lambda_{1k})} \neq 0\}$. Then, the k th site distributed its support estimates, $\hat{S}^{(k)}$, to all other sites except itself. Once all support estimates are exchanged, the k th site can construct a support union of all the other sites except site k , $\hat{S}^{(-k)} = \bigcup_{j \neq k} \hat{S}^{(j)}$, ensuring that the selected support $\hat{S}^{(-k)}$ is independent of the data at site k . There are two advantages: first, it reduces the probability of losing important features; second, it ensures the reliability of postselection inference in the next lower-dimensional estimation stage. Then, in the subsequent stage of estimating the lower-dimensional model, a straightforward approach would involve utilizing the standard maximum likelihood estimator (MLE) with the chosen covariates in $\hat{S}^{(-k)}$. However, in this paper, we do not employ MLE primarily for the reason that the MLE under GLMs may not have a unique solution. For example, in logistic regression, the MLE does not have a unique solution when the two classes are well-separated. Instead, we adopt a more modern approach by leveraging the refined desparsified Lasso method recently developed by [Xia et al. \(2023\)](#) for lower-dimensional GLMs with a diverging number of covariates. This approach first computes the Lasso estimator $\hat{\beta}_{\hat{S}^{(-k)}}^{(k,\lambda_{2k})}$ on the lower-dimensional model with covariates in $\hat{S}^{(-k)}$, where λ_{2k} is a tuning parameter selected via fivefold CV. Next, we debias this Lasso estimator to obtain the final refined desparsified Lasso estimator $\hat{\beta}_{\hat{S}^{(-k)}}^{(k,d)}$:

$$\hat{\beta}_{\hat{S}^{(-k)}}^{(k,d)} = \hat{\beta}_{\hat{S}^{(-k)}}^{(k,\lambda_{2k})} - \hat{\Theta}_{\hat{S}^{(-k)}} P_n \dot{L}_{\hat{\beta}_{\hat{S}^{(-k)}}^{(k,\lambda_{2k})}}, \quad (5)$$

where the estimated inverse Hessian matrix is given by $\hat{\Theta}_{\hat{S}^{(-k)}} = \hat{\Sigma}_{\hat{S}^{(-k)}}^{-1}$. In this setting, $\hat{\Sigma}_{\hat{S}^{(-k)}} = P_n \ddot{L}_{\hat{\beta}_{\hat{S}^{(-k)}}^{(k,\lambda_{2k})}}$ is directly invertible with probability approaching one under standard regularity conditions. In addition, [Xia et al. \(2023\)](#) showed that the refined desparsified Lasso outperforms both the original desparsified Lasso and the MLE in several simulation settings of lower-dimensional GLMs with sparse regression coefficients.

We summarize the complete Fed-FDR procedure in [Algorithm 2](#). The first part outlines the steps for obtaining the lower-dimensional refined desparsified Lasso estimator at each site k , detailing

Algorithm 2 Fed-FDR.

Require: $(\mathbf{Y}^{(k)}, \mathbf{X}^{(k)})$ for $1 \leq k \leq K$, and the given FDR level $\alpha \in (0, 1)$.
Ensure: The final support $\hat{S}$.

- 1: **Task for each site:**
- 2: **for** site $k \in \{1, \dots, K\}$ **do**
- 3: Fit GLM Lasso based on (4) to obtain the support $\hat{S}^{(k)}$ using the data $(\mathbf{Y}^{(k)}, \mathbf{X}^{(k)})$ with tuning parameter λ_{1k} selected by five-fold CV.
- 4: Distribute the support $\hat{S}^{(k)}$ to all the other sites except itself.
- 5: **end for**
- 6: **for** site $k \in \{1, \dots, K\}$ **do**
- 7: Construct support union $\hat{S}^{(-k)} = \bigcup_{j \neq k} \hat{S}^{(j)}$.
- 8: Obtain the estimator $\hat{\beta}_{\hat{S}^{(-k)}}^{(k,d)}$ based on (5), using the data $(\mathbf{Y}^{(k)}, \mathbf{X}^{(k)})$ with covariates restricted to $\hat{S}^{(-k)}$ and transfer $\hat{\beta}_{\hat{S}^{(-k)}}^{(k,d)}$ to the central site.
- 9: **end for**
- 10: **Task for the central site:**
- 11: Construct $\hat{\beta}^{(k)}$ by letting $\hat{\beta}_{\hat{S}^{(-k)}}^{(k)} = \hat{\beta}_{\hat{S}^{(-k)}}^{(k,d)}$ and $\hat{\beta}_{(\hat{S}^{(-k)})^c}^{(k)} = 0$ for $1 \leq k \leq K$.
- 12: Run [Algorithm 1](#) with $\hat{\beta}^{(k)}$ ($1 \leq k \leq K$) and α as input to obtain final support $\hat{S}$.

the individual tasks for each site. The second part involves the implementation of [Algorithm 1](#) to derive the final feature support. This entire procedure requires only a single round of communication, with the cost of transmitting vectors of dimension $|\hat{S}^{(-k)}|$ for each site. As shown in Section 4, this communication cost is of the order $O(Kp_1)$, representing the minimal communication overhead required for selecting a common feature set of size p_1 in the federated setting.

4. Theoretical results

For the data in each site, we make the following assumptions:

- A2 The covariates in $\mathbf{x}_{ki}$ are sub-Gaussian random variables for $1 \leq k \leq K$, $1 \leq i \leq n$.
A3 The design matrix $\Sigma_k = \mathbb{E}(\mathbf{x}_{ki}^T \mathbf{x}_{ki})$ as well as the weighted design matrix $\Sigma_{\beta^{(k*)}} = \mathbb{E}(P_n \tilde{L}_{\beta^{(k*)}})$ are positive definite, i.e. $1/M_1 < \lambda_{\min}(\Sigma_k) \leq \lambda_{\max}(\Sigma_k) < M_1$ and $1/M_2 < \lambda_{\min}(\Sigma_{\beta^{(k*)}}) \leq \lambda_{\max}(\Sigma_{\beta^{(k*)}}) < M_2$ for $1 \leq k \leq K$ and some constants $M_1, M_2 > 0$.
A4 The third-order derivative $\rho'''(x) = \partial^3 \rho(x) / \partial x^3$ is continuous.
A5 The sparsity p_1 satisfies $p_1 \log(p)/n = o(1)$ and the signal strength are large enough that $\min_{k \in [K], j \in S_1} \sigma_j^k / (p_1 \sqrt{\log(p)/n}) \rightarrow \infty$ as $n \rightarrow \infty$.

Assumptions A2 and A3 are typical in high-dimensional literature. Assumption A4 ensures that $\rho''(x)$ is Lipschitz continuous and that $\rho'(x)$ is bounded for $|x| \leq M$, where M is a sufficiently large constant. This condition holds for most commonly used GLMs, allowing the results for the refined desparsified Lasso estimator of [Xia et al. \(2023\)](#) to be applied. Assumption A5 guarantees that, with high probability, model selection based on the Lasso does not miss any relevant features, so that the selected model is not misspecified. A similar assumption also appears in data splitting procedures ([Dai et al., 2023a, 2023b](#); [Vazquez & Nan, 2025](#)) and high-dimensional variable selection procedures ([Bühlmann & Mandozzi, 2014](#); [Fan et al., 2014](#)) to guarantee the sure screening properties. This assumption is essential to ensure that the asymptotic power approaches one. However, if the signal strength σ_j^k is extremely small, specifically, if $\sigma_j^k \leq O(1/\sqrt{n})$ for some $j \in S_1$ and $k \in 1, \dots, K$, it becomes indistinguishable from the null features. In such cases, the problem shifts to a setting where the feature support exhibits slight heterogeneity across sites. We provide a detailed discussion of this scenario in Section 4.1.

Proposition 2 ((Variable screening properties for the GLM Lasso)). For a choice of tuning parameter $\lambda_{1k} = O(\sqrt{\log(p)/n})$ and under assumptions A1-A5, the Lasso estimator $\hat{\beta}^{(k, \lambda_{1k})}$ and its selected model $\hat{S}^{(k)} = \{j : \hat{\beta}_j^{(k, \lambda_{1k})} \neq 0\}$ satisfy

$$P\left(S_1 \subseteq \hat{S}^{(k)}, |\hat{S}^{(k)}| \leq c_0 p_1\right) \geq 1 - c_1/p^{c_2}$$

for some positive constants c_0, c_1, c_2 .

Proposition 2 demonstrates that under mild conditions, the Lasso selects a model containing the true signal variables with high probability. However, a large number of noise variables of the order $O(p_1)$ still remain unscreened. Therefore, it is crucial to utilize generalized mirror statistics to screen out these noise variables while controlling the FDR. On the other hand, the result implies that the Lasso selected model size is of the same order as the number of nonzero regression coefficients, with high probability. This is essential because it ensures that the local site only needs to transfer a vector of lower dimensional coefficients that significantly reduces the communication cost. In addition, if the selected model size is too large then its design matrix may not have full rank, even asymptotically, and thus the regression coefficients will still not be estimable via MLE or even the refined desparsified Lasso of Xia et al. (2023).

Proposition 3 ((Asymptotic normality)). For the desparsified Lasso estimator $\hat{\beta}_{\hat{S}^{(k)}}^{(k, d)}$ with tuning parameters $\lambda_{2k} = O(\sqrt{\log(p_1)/n})$, under assumptions A1-A5 and further assume that $p_1^{3/2} \log(p_1) = o(\sqrt{n})$, $Kp_1 = o(\sqrt{n})$, we have $\hat{\Sigma}_{\hat{S}^{(k)}}$ is invertible with probability going to one. Furthermore, conditional on the event that $S_1 \subseteq \hat{S}^{(k)}$ occurs with probability approach to one (as established in Proposition 2) and noting that $\hat{S}^{(k)}$ is independent of the data at site k , we have

$$\sqrt{n}(\hat{\beta}_{\hat{S}^{(k)}}^{(k, d)} - \beta_{\hat{S}^{(k)}}^{(k*)}) \rightarrow N(0, \hat{\Theta}_{\hat{S}^{(k)}})$$

in distribution as $n \rightarrow \infty$.

Proposition 3 establishes that the refined desparsified Lasso estimator within the lower-dimensional support set follows an asymptotic normal distribution. This result guarantees that the distribution of the selected null features becomes (asymptotically) symmetric around zero, thereby satisfying the symmetry requirement of the coefficient estimates. Moreover, as demonstrated in Lemma 7 of the online supplementary material, the generalized mirror statistics constructed using the refined desparsified Lasso estimator also satisfy the weak dependency condition. Then, by Proposition 2, we have the following Theorem 1 shows the asymptotic FDR control and the power of the proposed Fed-FDR method.

Theorem 1 For any FDR control level $\alpha \in (0, 1)$, under the same assumptions as in Proposition 3, we have

- (a) $\text{FDP} \leq \alpha + O_p\left(\frac{1}{nK}\right)$ and $\limsup_{nK, p \rightarrow \infty} \text{FDR} \leq \alpha$,
 - (b) $\liminf_{nK, p \rightarrow \infty} \text{Power} = 1$,
- for the federated feature selection procedure summarized in Algorithms 2.

From Theorem 1(a), we observe that the FDP is controlled at a faster rate as the number of sites and the local sample size increases, therefore, the FDR can be maintained at the desired level α . Furthermore, Theorem 1(b) demonstrates that the asymptotic power of the proposed approach is equal to one.

4.1 Support heterogeneity across sites

In Assumption A1, we assume that $\beta_j^{(k*)} = \sigma_j^k \beta_j$, where the signal strength σ_j^k varies across sites, while the underlying feature support $S_1 = \{j \in \{1, \dots, p\} | \beta_j \neq 0\}$ remains the same across all sites.

However, as more sites are incorporated, ensuring perfect overlap in feature support across all sites becomes challenging. For example, consider a multi-site study investigating risk factors influencing GLP-1 medication adherence in type 2 diabetes patients. Differences in patient population characteristics across healthcare institutions may lead to site-specific risk factors, such as variations in the impact of insurance type or health awareness. However, factors like age, number of comorbidities, and follow-up frequency are likely to be consistently important across all sites. In this setting, where different sites have slightly varying feature supports, the primary interest is often in identifying the overlap, i.e. the intersection of all feature supports, which represents the common set of important features across sites. Additionally, we also investigate the case of identifying the union of all feature supports, accommodating scenarios where one may be interested in capturing all relevant features present in at least one site.

In this section, we extend the homogeneous support setting to a more general framework that allows for mild heterogeneity in feature support across sites, and evaluate the performance of our proposed method in relation to the two objectives discussed above. Specifically, for $k = 1, \dots, K$, denote $S_1^{(k)} = \{j \in \{1, \dots, p\} | \beta_j^{(k*)} \neq 0\}$ be the feature support with respect to site k . Let $S_1 = \bigcap_{1 \leq k \leq K} S_1^{(k)}$ denote the set of relevant features shared across all sites, and let $U_1 = \bigcup_{1 \leq k \leq K} S_1^{(k)}$ represent the union of all relevant features present in at least one site. If we focus on identifying S_1 , the corresponding FDR and power are given by $\text{FDR}_{S_1} = \mathbb{E}(\text{FDP}_{S_1})$ with $\text{FDP}_{S_1} = \frac{|\hat{S}_1 \cap S_1^c|}{|\hat{S}_1|}$ and $\text{Power}_{S_1} = \frac{\mathbb{E}(|\hat{S}_1 \cap S_1|)}{|\hat{S}_1|}$, respectively. It is obvious that the features belonging to S_1 that have been previously selected in the homogeneous setting can also be selected in the heterogeneous setting, ensuring no loss of power. However, for features in $U_1 \setminus S_1$, i.e. those that belong to U_1 but not S_1 , their inclusion rate is higher than that of null features, leading to an increased FDR. On the contrary, the FDR can be controlled if the goal is to identify U_1 , where the corresponding FDR and power are given by $\text{FDR}_{U_1} = \mathbb{E}(\text{FDP}_{U_1})$ with $\text{FDP}_{U_1} = \frac{|\hat{S}_1 \cap U_1^c|}{|\hat{S}_1|}$ and $\text{Power}_{U_1} = \frac{\mathbb{E}(|\hat{S}_1 \cap U_1|)}{|\hat{S}_1|}$, respectively. For features that belong to $U_1 \setminus S_1$, their inclusion rate is lower than that of nonnull features in S_1 , leading to a loss of power. We summarize the properties in the following theorem.

Theorem 2 For any FDR control level $\alpha \in (0, 1)$, under Assumptions A2–A5 and the heterogeneous support setting. There exists a constant $N^* = c(S_1^{(1)}, \dots, S_1^{(K)}, \alpha)$, satisfying $|S_1| \leq N^* \leq |U_1|$, such that the federated feature selection procedure summarized in [Algorithm 2](#) satisfies the following:

$$\limsup_{nK, p \rightarrow \infty} \text{FDR}_{S_1} \leq \alpha + (1 - \alpha) \frac{N^* - |S_1|}{|U_1|} \quad \text{and} \quad \liminf_{nK, p \rightarrow \infty} \text{Power}_{S_1} = 1,$$

when identifying S_1 . If we instead consider identifying U_1 , we obtain

$$\limsup_{nK, p \rightarrow \infty} \text{FDR}_{U_1} \leq \alpha \quad \text{and} \quad \liminf_{nK, p \rightarrow \infty} \text{Power}_{U_1} \geq 1 - \frac{|U_1| - N^*}{|U_1|}.$$

The function $c(S_1^{(1)}, \dots, S_1^{(K)}, \alpha)$, determined by all the support sets $S_1^{(k)}$ ($k = 1, \dots, K$) and the FDR level α , is provided in the proof of [Theorem 2 in the online supplementary material](#). Generally speaking, if the features in $U_1 \setminus S_1$ are sparsely distributed across all sites, meaning that only a few sites contain these features, then N^* will be close to $|S_1|$. In the extreme case where the features in $U_1 \setminus S_1$ appear in only one of the K sites, we have $N^* = |S_1|$. Conversely, if the features in $U_1 \setminus S_1$ are densely distributed across all sites, meaning they appear in most sites, then N^* will be close to $|U_1|$. In the extreme case where all features in $U_1 \setminus S_1$ appear in $K - 1$ of the K sites, we have $N^* = |U_1|$. In summary, under the heterogeneous support setting, the proposed approach effectively detects all common features shared across sites. However, the FDR and power depend on the specific task being considered. Moreover, in the setting with slight heterogeneity, where the support heterogeneity level satisfies

$\frac{|U_1|-|S_1|}{|U_1|} = o(1)$, we have both $\frac{N^*-|S_1|}{|U_1|} = o(1)$ and $\frac{|U_1|-N^*}{|U_1|} = o(1)$. Then, the result in Theorem 2 reduces to that of the homogeneous support setting as stated in Theorem 1.

4.2 Model misspecification

In the previous sections, we assume that y_{ki} , given the covariate $\mathbf{x}_{ki}$, follows a correctly specified GLM. In this section, we extend the proposed approach to scenarios where the true probability distribution $\mathbb{P}^*(y_{ki}, \mathbf{x}_{ki})$ does not belong to the assumed exponential family, while still using the GLM as a working model for estimation. In this setting, the parameter of interest, $\boldsymbol{\beta}^{(k*)}$, is defined as the minimizer of the Kullback–Leibler (KL) divergence (Lv & Liu, 2014), i.e.

$$\boldsymbol{\beta}^{(k*)} = \operatorname{argmin}_{\boldsymbol{\beta}} \text{KL}^{(k)}(\boldsymbol{\beta}) \quad \text{with } \text{KL}^{(k)}(\boldsymbol{\beta}) = \mathbb{E}^* \left\{ \log \frac{f^*(y_{ki}, \mathbf{x}_{ki})}{f(y_{ki}, \mathbf{x}_{ki}; \boldsymbol{\beta})} \right\}, \quad (6)$$

where $f^*(y_{ki}, \mathbf{x}_{ki})$ is the true density function of $(y_{ki}, \mathbf{x}_{ki})$, and $f(y_{ki}, \mathbf{x}_{ki}; \boldsymbol{\beta})$ is the density corresponding to the specified GLM working model with respect to parameter $\boldsymbol{\beta}$. Here, $\mathbb{E}^*(\cdot)$ denotes the expectation with respect to the true density function $f^*(y_{ki}, \mathbf{x}_{ki})$. Since minimizing the KL divergence (6) is equivalent to solving the score equation

$$\mathbb{E}^*\{y_{ki} - \rho'(\mathbf{x}_{ki}^\top \boldsymbol{\beta})\} \mathbf{x}_{ki} = 0. \quad (7)$$

Under the Quasi-Maximum Likelihood Estimation (QMLE) framework (Demirkaya et al., 2022; Ning & Liu, 2017), parameter estimates remain asymptotically consistent provided that $\boldsymbol{\beta}^{(k*)}$ satisfies the score equation in (7). Consequently, all the results in Theorem 1 remain valid provided that $\boldsymbol{\beta}^{(k*)}$ fulfils this condition. This allows our approach to serve as a ‘working model’ that facilitates estimation while remaining robust to certain forms of model misspecification.

5. Simulation

In this section, we investigate the finite sample performance of the proposed Fed-FDR method and compare it with some benchmark methods. For simplicity, we use Fed-FDR to refer to the proposed framework in Algorithm 2. To examine the impact of the variable selection method in Step 3 of Algorithm 2, we also consider Fed-FDR-SCAD, which replaces only Step 3 with SCAD penalty proposed by Fan and Li (2001) for variable selection at each local site. When the data cannot be shared, a straightforward approach involves adopting a concept similar to that of Cai et al. (2022) for estimation and constructing p-values for each feature, followed by the utilization of the BHq method for feature selection with FDR control, which we refer to as Meta-BHq. Another benchmark approach involves each local site first constructing its own desparsified Lasso estimator $\hat{\beta}^{(k,d)}$ with the decorrelating matrix $\hat{\Theta}_k$ calculated with a node-wise Lasso procedure. Then, Algorithm 1 is executed using the collected $\hat{\beta}^{(k,d)}$ values from each site as input to obtain the final feature support, which we called Fed-FDR-N. The detailed steps of the Meta-BHq and Fed-FDR-N methods are provided in Section S.2 of the online supplementary material. While the last two methods prioritize privacy protection, it does require each site to transfer the de-sparsity estimator, which has the order of $O(Kp)$. When the data can be pooled together, we consider the BHq method as the ‘gold standard’. This approach first obtains p-values using desparsified Lasso (Van de Geer et al., 2014) and then applies the standard BHq procedure proposed by Benjamini and Hochberg (1995).

We consider GLMs, the response $y_{ki} \sim \text{Bionomial}(\pi_{ki})$, which is generated from a logistic regression with $\pi_{ki} = \text{expit}(\gamma_k + \mathbf{x}_{ki}^\top \boldsymbol{\beta}^{(k*)})$, and $\text{expit}(x) = 1/(1 + \exp(-x))$. For all synthetic examples, we set $|\beta_j^{(k*)}| = \sigma_j^k \sqrt{\log(p)/n}$ for the relevant features $j \in S_1$ and randomly generate their signs with equal probability. We refer to $\sigma_j^k > 0$ as the signal strength for the j th feature in the k th site. The p -dimensional features are generated from $\mathbf{x}_{ki} \sim N(0, \Sigma_k)$ with the Toeplitz correlation structure, that is, $\Sigma_{k;st} = \rho_k^{|s-t|}$. The designated FDR control level is set to $\alpha = 0.1$ henceforth. Each value of the dot in the figures represents the average from 50 independent runs.

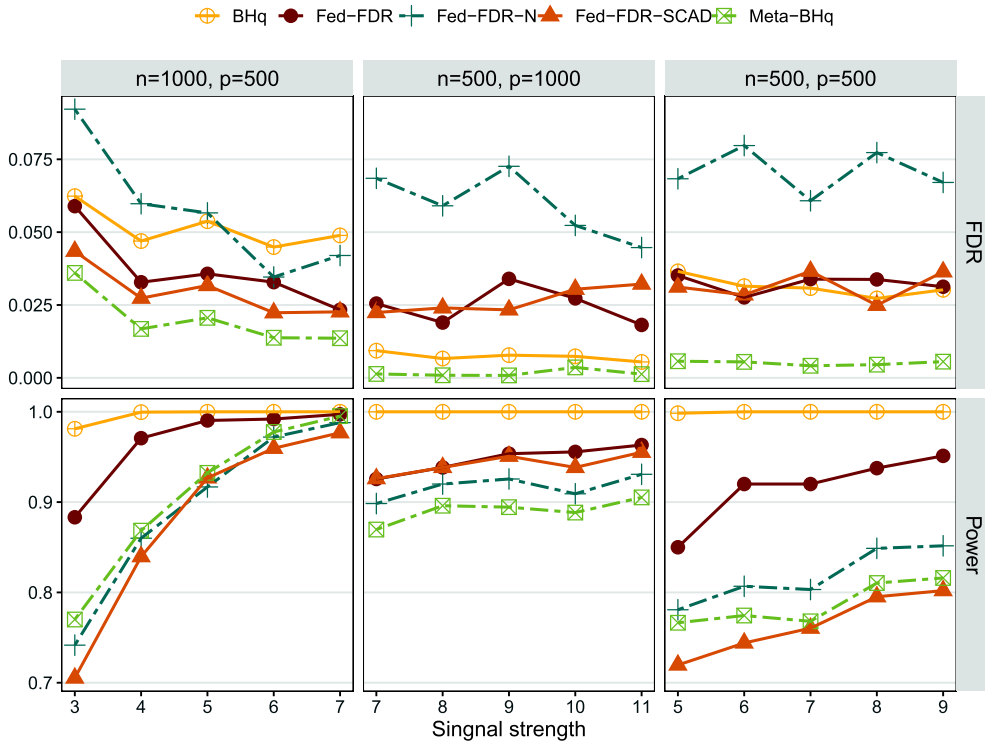


Figure 1. Comparison of empirical false discovery rates (FDRs) and power as a function of signal strength across different combinations of local sample size n and parameter dimension p . The number of relevant features is $p_1 = 50$, with a total of $K = 5$ sites. The parameters $\beta^{(k*)}$ and ρ_k are assumed to be identical across sites. Without loss of generality, we set $\gamma_k = -0.5$ for $k \in \{1, \dots, K\}$ and fix the correlation at $\rho_k = 0.5$ for all sites. Performance is evaluated by varying the signal strength.

We first evaluate the performance of the proposed approach by varying the signal strength. Specifically, we consider the case where the number of relevant features is $p_1 = 50$, and there are a total of $K = 5$ sites. Without loss of generality, we set $\gamma_k = -0.5$ and $\rho_k = 0.5$ for $k \in \{1, \dots, K\}$. The comparison results of empirical FDRs and powers for different methods, as a function of signal strength across various combinations of local sample size n and parameter p , are shown in Figure 1. The top three subfigures demonstrate that the FDRs of all five competing methods are effectively controlled across different settings. Regarding power, the proposed Fed-FDR method outperforms the Meta-BHq, Fed-FDR-N, and Fed-FDR-SCAD methods in nearly all scenarios. Additionally, when comparing Fed-FDR with Fed-FDR-N, we observe that the Fed-FDR methods achieve higher powers in almost all scenarios while only requiring the sharing of a lower-dimensional vector of coefficients for each site, thus significantly reducing communication costs in real-world applications. In contrast, the power of the Fed-FDR-SCAD method is relatively low when $p = 500$. This could be due to the SCAD penalty's tendency to select fewer variables than Lasso, which may cause a loss of power.

We then evaluate the performance of the proposed approach by varying the correlation of the covariates. Specifically, we consider the case where the number of relevant features is $p_1 = 50$, and there are a total of $K = 5$ sites. Additionally, we set $\gamma_k = -0.5$ and fix the signal strength at $\sigma_j^k = 4$, $\sigma_j^k = 9$, and $\sigma_j^k = 7$ for the left, middle, and right panels, respectively. The comparison results of empirical FDRs and powers for different methods, as a function of correlation ρ_k across various combinations of local sample size n and parameter p , are shown in Figure 2. It can be seen from Figure 2 that the proposed Fed-FDR method outperforms all other methods in nearly all scenarios, except for the BHq method, which pools all the data.

We finally examine the performance in scenarios involving variations in local sample size n , number of sites K , and sparsity level p_1 , respectively. For all three cases, the parameters are set

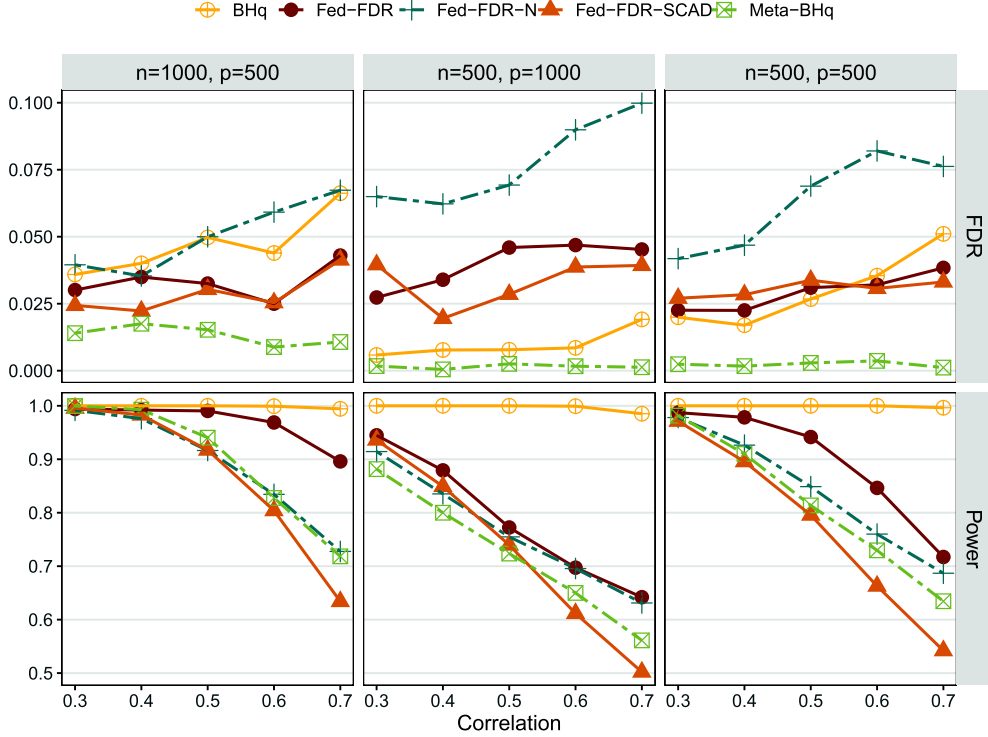


Figure 2. Comparison of empirical false discovery rates (FDRs) and power as a function of correlation ρ_k across different combinations of local sample size n and parameter dimension p . The number of relevant features is $p_1 = 50$, with a total of $K = 5$ sites. The parameters $\boldsymbol{\mu}^{(k*)}$ and ρ_k are assumed to be identical across sites. Without loss of generality, we set $\gamma_k = -0.5$ for $k \in \{1, \dots, K\}$ and fix the signal strength at $\sigma_j^k = 4$, $\sigma_j^k = 9$, and $\sigma_j^k = 7$ for the left, middle, and right panels, respectively. Performance is evaluated by varying the correlation ρ_k .

as $\gamma_k \sim \text{Uniform}(-0.5, 0.5)$ and $\rho_k \sim \text{Uniform}(0.3, 0.5)$, varying across sites. The signal strength is set to $\sigma_j^k \sim \text{Uniform}(7, 11)$ for $j \in 1, \dots, p$ and $k \in 1, \dots, K$. The comparison results are presented in Figure 3. In the left panel, the number of relevant features is fixed at $p_1 = 50$ and performance is evaluated by varying the number of local sample size n . In the middle panel, we fix the number of relevant features at $p_1 = 50$ and the local sample size at $n = 500$, evaluating performance by varying the number of sites K . In the right panel, the local sample size and site number are fixed at $n = 500$ and $K = 5$, and performance is evaluated by varying the number of relevant features p_1 . These three panels demonstrate that the proposed Fed-FDR method outperforms other methods, except for BHq, across almost all settings in terms of power. As the local sample size or the number of sites increases, the power of the proposed approach also increases rapidly, approaching one at a fast rate. Moreover, to assess whether the proposed method is significantly influenced by the tuning parameter λ_{1k} , we compare the performance of the Fed-FDR and Fed-FDR-SCAD methods under different scalings of the optimal λ_{1k} chosen by fivefold CV. The scaling factor is set to $\{0.5, 1, 1.5, 2, 2.5\}$. The comparison results are presented in Figure S.1 of the online supplementary material, which demonstrates that the proposed method is more robust to variations in the tuning parameters.

Overall, the extensive simulation studies demonstrate that the proposed method is robust to the heterogeneity of the feature distribution, easy to implement, and computationally efficient. Additionally, this framework successfully maintains FDR control while achieving competitive statistical power.

6. Applications

In recent years, several studies have investigated acute sequelae of SARS-CoV-2 infection (PASC) among adults infected with SARS-CoV-2 (Hill et al., 2023; Proal & VanElzakker, 2021; Wang

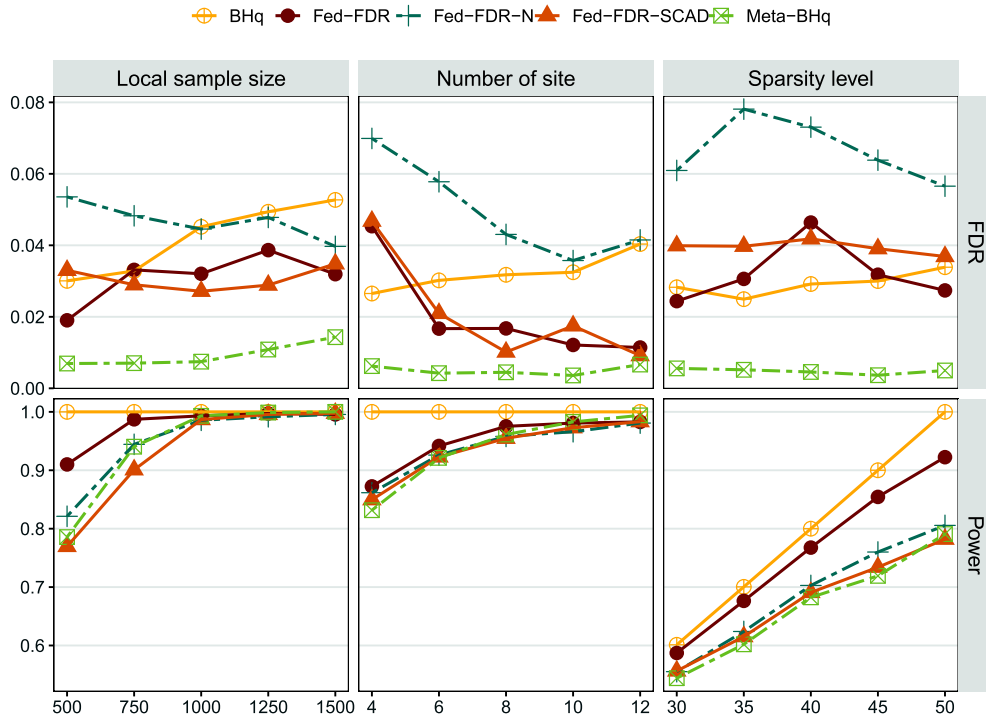


Figure 3. Comparison of empirical false discovery rates (FDRs) and power as a function of local sample size n (left panel), number of site K (middle panel), and sparsity level p_1 (right panel). The parameters are set as $p = 500$, $\gamma_k \sim \text{Uniform}(-0.5, 0.5)$, $\rho_k \sim \text{Uniform}(0.2, 0.8)$, and $\sigma_j^k \sim \text{Uniform}(7, 11)$ for each site, with a total of $K = 5$ sites. In the left panel, the number of relevant features is fixed at $p_1 = 50$, and performance is assessed by varying the number of local sample size n . In the middle panel, we fix the number of relevant features at $p_1 = 50$ and the local sample size at $n = 500$, evaluating performance by varying the number of site K . In the right panel, the local sample size and site number are fixed at $n = 500$ and $K = 5$, and performance is evaluated by varying the number of relevant features p_1 .

et al., 2023; Zang et al., 2023; Zhang et al., 2023). However, research on PASC symptoms and conditions in pediatric populations in the United States, particularly in identifying key features related to clinical outcomes, remains limited. In this section, we illustrate the proposed methods utilizing a simulated data set generated from the summary statistics derived from EHR data in existing studies (Thaweethai et al., 2023; Wu et al., 2024). The objective is to investigate the important features associated with PASC in children during the Omicron period.

The eligibility criteria for participants in this cohort included being between the ages 5–11 at the beginning of the study period, with no prior COVID-19 vaccination or documented SARS-CoV-2 infection. Additionally, participants were required to have had a prior encounter (including telephone or telehealth encounters) within 18 months before entering the cohort to ensure ongoing interaction with the healthcare system. The outcome of interest was the occurrence of the most prevalent PASC features in our data—specifically, allergic reactions—observed within 28–179 days following the initial SARS-CoV-2 test date. The FDR level is set to be $\alpha = 0.1$ for all the comparison methods.

To mimic the patients' medical records in the EHR data, we simulated a set of confounders using the summary statistics reported in the existing studies, including demographic variables such as age, sex, race, and obesity status. We also simulated the Pediatric Medical Complexity Algorithm (PMCA) score (Simon et al., 2018), vaccination status during the study period, the number of emergency department visits in the 18 months leading up to 7 days before cohort entry, the number of inpatient visits during the same time frame, the number of outpatient visits within the specified period, the count of unique medications prescribed within the 18 months prior to 7 days before cohort entry, and the presence of diagnoses related to 205 chronic condition clusters within the same time frame. In total, we included 243 binary features in our analysis.

Table 1. Feature selected by different methods

Features	BHq	Fed-FDR-N	Meta-BHq	Fed-FDR-SCAD	Fed-FDR
History of obesity	✓	✓	✓	✓	✓
History of allergies	✓	✓	✓	✓	✓
History of asthma	✓			✓	✓
History of psychological symptoms	✓	✓	✓		
History of seizures and epilepsy	✓	✓	✓		
Race—White	✓				
Cohort Entry Month—12,2021	✓			✓	✓
Number of inpatient visits (=0)	✓				
Race - Black				✓	
History of anxiety disorder		✓	✓		
History of acquired foot deformities		✓			
History of blindness & vision defects		✓			
History of autism spectrum disorder		✓			
History of cornea and external disease		✓			
History of oesophageal disorders		✓			
History of developmental delay		✓			
Number of emergency department visits (2 or more)		✓	✓		
Number of inpatient visits (=1)		✓	✓		
Number of outpatient visits (2 or more)		✓			✓
Number of medications prescribed (2 or more)				✓	

Note. BHq = BenjaminHochberg; FDR = false discovery rate.

Our final simulated dataset consisted of 3,990 individuals collected from 34 clinical sites, with each site contributing approximately 130 patients. We had complete access to all 3,990 individuals, allowing us to apply various methods, including the BHq method (considered the gold standard, as it is typically most powerful when data can be pooled together, as evidenced by numerous simulations), as well as Fed-FDR-N, Meta-BHq, Fed-FDR (i.e. proposed method), and Fed-FDR-SCAD. As illustrated in Table 1, the BHq method identified eight features in total. Notably, four of these features were also identified by the proposed Fed-FDR method. While the Fed-FDR-SCAD, Fed-FDR-N, and Meta-BHq methods each identified four of these eight features, they also selected additional features that were not supported by the gold standard method, suggesting a less reliable selection performance.

The features identified by the gold standard method and our proposed methods have been found to be significant risk factors for PASC in existing studies of the adult population. For instance, preexisting conditions such as obesity, allergies, and anxiety disorder have been shown to be associated with an increased risk of PASC (Hill et al., 2023; Jacobs et al., 2023; Zang et al., 2023). However, the populations studied in these references differ from pediatric populations. In this study, we focused only on one of the PASC conditions for the pediatric population. Therefore, there is a pressing need for further investigation beyond the current analysis into PASC risk factors specifically among children to understand how these factors may uniquely impact the younger demographic.

We further compared the predictive power of the existing and proposed methods. The data was randomly divided within each site, allocating 80% for training purposes and 20% for testing, and we compared the receiver operating characteristics (ROC) curves of different methods. As shown in Figure 4, the area under the ROC curve (AUC) for the proposed method (i.e. Fed-FDR) was closer to that of the gold standard method and greater than those of the Meta-BHq and Fed-FDR-N

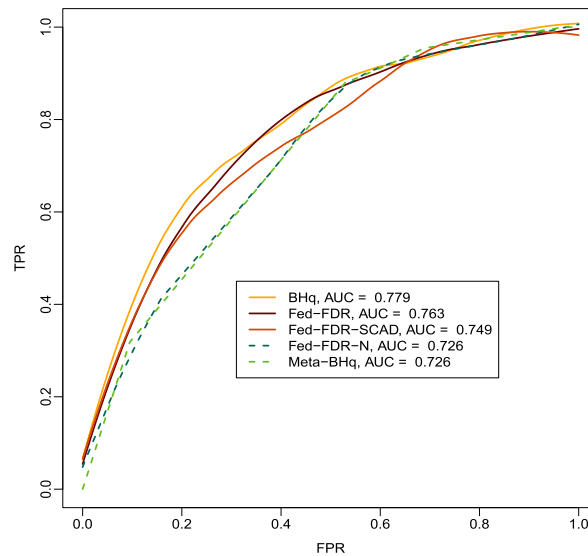


Figure 4. Receiver operating characteristics (ROC) curves for Benjamin-Hochberg (BHq), Fed-FDR-N, Meta-BHq, Fed-FDR-SCAD, and Fed-FDR (our proposed) methods.

methods. Specifically, the AUCs were 0.779 for BHq, 0.763 for Fed-FDR, 0.749 for Fed-FDR-SCAD, 0.726 for Fed-FDR-N, and 0.726 for Meta-BHq. This result suggests that, within the pediatric cohort we analysed, our Fed-FDR method identified risk factors that appeared to be more predictive compared to the Meta-BHq, Fed-FDR-N, and Fed-FDR-SCAD methods. However, further investigation and evaluation are needed to evaluate these results using other databases.

7. Concluding remarks

We have introduced a comprehensive framework for performing federated feature selection in GLM, with the FDR asymptotically under control. Specifically, we present the Fed-FDR method, which requires each site to compute lower-dimensional coefficient estimates based on the support union derived from other sites. These estimates are then transferred to the central site, where generalized mirror statistics are constructed for feature selection. We show that the communication cost of Fed-FDR is proportional to the number of relevant features, which represents the minimum communication cost achievable for federated feature selection. Both our simulation study and real data application demonstrate that the proposed methods outperform the Fed-FDR-N method, which requires node-wide Lasso, and the meta-BHq method, which relies on p-values, in terms of power, communication cost, and computational efficiency. Furthermore, we show that Fed-FDR is also more robust to tuning parameters compared to the Fed-FDR-SCAD method.

In this paper, our emphasis lies in minimizing communication costs and safeguarding private data information, crucially considerations in real-world applications. To achieve these objectives, we introduce the GLM model and employ generalized mirror statistics to obtain the desired results. However, when operating under more relaxed constraints, such as sharing high dimensional $p \times 1$ vectors, a model-free framework that base on a certain correlation distance measure (Liu et al., 2022) or conditional correlation distance measure (Tong et al., 2023) can be employed for federated feature selection via FDR control, and these types of directions could be interesting in future work.

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Conflicts of interest: The authors do not declare any conflict of interest regarding this work.

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Supplementary material

Supplementary material is available online at *Journal of the Royal Statistical Society: Series B*.

Data availability

The synthetic data is available at our GitHub repository: <https://github.com/Penncil/Fed-FDR>.

Code availability

The source code for the Fed-FDR method is freely available for academic use at GitHub: <https://github.com/Penncil/Fed-FDR>.

References

- Barber R. F., & Candès E. J. (2015). Controlling the false discovery rate via knockoffs. *Annals of Statistics*, 43(5), 2055–2085. <https://doi.org/10.1214/15-AOS1337>
- Batthey H., Fan J., Liu H., Lu J., & Zhu Z. (2018). Distributed testing and estimation under sparse high dimensional models. *Annals of Statistics*, 46(3), 1352. <https://doi.org/10.1214/17-AOS1587>
- Behlen F. M., & Johnson S. B. (1999). Multicenter patient records research: Security policies and tools. *Journal of the American Medical Informatics Association*, 6(6), 435–443. <https://doi.org/10.1136/jamia.1999.0060435>
- Benjamini Y., & Hochberg Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Benjamini Y., & Yekutieli D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188. <https://doi.org/10.1214/aos/1013699998>
- Bühlmann P., & Mandozzi J. (2014). High-dimensional variable screening and bias in subsequent inference, with an empirical comparison. *Computational Statistics*, 29(3–4), 407–430. <https://doi.org/10.1007/s00180-013-0436-3>
- Cai T., Liu M., & Xia Y. (2022). Individual data protected integrative regression analysis of high-dimensional heterogeneous data. *Journal of the American Statistical Association*, 117(540), 2105–2119. <https://doi.org/10.1080/01621459.2021.1904958>
- Candès E., Fan Y., Janson L., & Lv J. (2018). Panning for gold: ‘Model-x’ knockoffs for high dimensional controlled variable selection. *Journal of the Royal Statistical Society Series: Series B (Methodological)*, 80(3), 551–577. <https://doi.org/10.1111/rssb.12265>
- Cao Y., Sun X., & Yao Y. (2024). Controlling the false discovery rate in transformational sparsity: Split knockoffs. *Journal of the Royal Statistical Society: Series B, Statistical Methodology*, 86(2), 386–410. <https://doi.org/10.1093/jrsssb/qkad126>
- Dai C., Lin B., Xing X., & Liu J. S. (2023a). False discovery rate control via data splitting. *Journal of the American Statistical Association*, 118(544), 2503–2520. <https://doi.org/10.1080/01621459.2022.2060113>
- Dai C., Lin B., Xing X., & Liu J. S. (2023b). A scale-free approach for false discovery rate control in generalized linear models. *Journal of the American Statistical Association*, 118(543), 1551–1565. <https://doi.org/10.1080/01621459.2023.2165930>
- Demirkaya E., Feng Y., Basu P., & Lv J. (2022). Large-scale model selection in misspecified generalized linear models. *Biometrika*, 109(1), 123–136. <https://doi.org/10.1093/biomet/asab005>
- Fan J., Fan Y., & Barut E. (2014). Adaptive robust variable selection. *Annals of Statistics*, 42(1), 324. <https://doi.org/10.1214/13-AOS1191>
- Fan J., & Li R. (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *Journal of the American Statistical Association*, 96(456), 1348–1360. <https://doi.org/10.1198/016214501753382273>
- Hill E., Mehta H., Sharma S., Mane K., Xie C., Cathey E., Lommba J., Russell S., Spratt H., DeWitt P. E., et al. (2023). Risk factors associated with post-acute sequelae of SARS-CoV-2: an N3C and NIH RECOVER study. *BMC Public Health*, 23(1), 2103. <https://doi.org/10.1186/s12889-023-16916-w>
- Huang D., & Janson L. (2020). Relaxing the assumptions of knockoffs by conditioning. *Annals of Statistics*, 48(5), 3021–3042. <https://doi.org/10.1214/19-AOS1920>

- Jacobs E. T., Catalfamo C. J., Colombo P. M., Khan S. M., Austhof E., Cordova-Marks F., Ernst K. C., Farland L. V., & Pogreba-Brown K. (2023). Pre-existing conditions associated with post-acute sequelae of COVID-19. *Journal of Autoimmunity*, 135(8), 102991. <https://doi.org/10.1016/j.jaut.2022.102991>
- Javanmard A., & Javadi H. (2019). False discovery rate control via debiased lasso. *Electronic Journal of Statistics*, 13(1), 1212–1253. <https://doi.org/10.1214/19-EJS1554>
- Kumar V., & Minz S. (2014). Feature selection: A literature review. *SmartCR*, 4(3), 211–229. <https://doi.org/10.6029/smartcr.2014.03.007>
- Lee J. D., Liu Q., Sun Y., & Taylor J. E. (2017). Communication-efficient sparse regression. *Journal of Machine Learning Research: JMLR*, 18(1), 115–144. <http://jmlr.org/papers/v18/16-002.html>
- Liu M., Xia Y., Cho K., & Cai T. (2021). Integrative high dimensional multiple testing with heterogeneity under data sharing constraints. *Journal of Machine Learning Research: JMLR*, 22(1), 5607–5632. <http://jmlr.org/papers/v22/20-774.html>
- Liu W., Ke Y., Liu J., & Li R. (2022). Model-free feature screening and FDR control with knockoff features. *Journal of the American Statistical Association*, 117(537), 428–443. <https://doi.org/10.1080/01621459.2020.1783274>
- Lv J., & Liu J. S. (2014). Model selection principles in misspecified models. *Journal of the Royal Statistical Society: Series B, Statistical Methodology*, 76(1), 141–167. <https://doi.org/10.1111/rssb.12023>
- Ma R., Tony Cai T., & Li H. (2021). Global and simultaneous hypothesis testing for high-dimensional logistic regression models. *Journal of the American Statistical Association*, 116(534), 984–998. <https://doi.org/10.1080/01621459.2019.1699421>
- Miao J., & Niu L. (2016). A survey on feature selection. *Procedia Computer Science*, 91(04), 919–926. <https://doi.org/10.1016/j.procs.2016.07.111>
- Ning Y., & Liu H. (2017). A general theory of hypothesis tests and confidence regions for sparse high dimensional models. *Annals of Statistics*, 47(6), 158–195. <https://doi.org/10.1214/16-AOS1448>
- Proal A. D., & VanElzakker M. B. (2021). Long COVID or post-acute sequelae of COVID-19 (PASC): An overview of biological factors that may contribute to persistent symptoms. *Frontiers in Microbiology*, 12, 698169. <https://doi.org/10.3389/fmicb.2021.698169>
- Sarkar S. K., & Tang C. Y. (2022). Adjusting the Benjamini–Hochberg method for controlling the false discovery rate in knockoff-assisted variable selection. *Biometrika*, 109(4), 1149–1155. <https://doi.org/10.1093/biomet/asab066>
- Simon T. D., Haaland W., Hawley K., Lambka K., & Mangione-Smith R. (2018). Development and validation of the Pediatric Medical Complexity Algorithm (PMCA) version 3.0. *Academic Pediatrics*, 18(5), 577–580. <https://doi.org/10.1016/j.acap.2018.02.010>
- Thaweethai T., Jolley S. E., Karlson E. W., Levitan E. B., Levy B., McComsey G. A., McCorkell L., Nadkarni G. N., Parthasarathy S., Singh U., Walker T. A., Selvaggi C. A., Shinnick D. J., Schulte C. C. M., Atchley-Challenor R., Horwitz L. I., & Foulkes A. S. (2023). Development of a definition of postacute sequelae of SARS-CoV-2 infection. *JAMA*, 329(22), 1934–1946. <https://doi.org/10.1001/jama.2023.8823>
- Tong Z., Cai Z., Yang S., & Li R. (2023). Model-free conditional feature screening with FDR control. *Journal of the American Statistical Association*, 118(544), 2575–2587. <https://doi.org/10.1080/01621459.2022.2063130>
- Van de Geer S., Bühlmann P., Ritov Y., & Dezeure R. (2014). On asymptotically optimal confidence regions and tests for high-dimensional models. *Annals of Statistics*, 42(3), 1166–1202. <https://doi.org/10.1214/14-AOS1221>
- Vazquez O., & Nan B. (2025). Debiased lasso after sample splitting for estimation and inference in high dimensional generalized linear models. *Canadian Journal of Statistics*, 53(1), e11827. <https://doi.org/10.1002/cjs.11827>
- Wang C., Ramasamy A., Verduzco-Gutierrez M., Brode W. M., & Melamed E. (2023). Acute and post-acute sequelae of SARS-CoV-2 infection: A review of risk factors and social determinants. *Virology Journal*, 20(1), 124. <https://doi.org/10.1186/s12985-023-02061-8>
- Wu Q., Tong J., Zhang B., Zhang D., Chen J., Lei Y., Lu Y., Wang Y., Li L., Shen Y., Xu J., Bailey L. C., Bian J., Christakis D. A., Fitzgerald M. L., Hirabayashi K., Jhaveri R., Khaitan A., Lyu T., ...Chen Y. (2024). Real-world effectiveness of BNT162b2 against infection and severe diseases in children and adolescents. *Annals of Internal Medicine*, 177(2), 165–176. <https://doi.org/10.7326/M23-1754>
- Xia L., Nan B., & Li Y. (2023). Debiased lasso for generalized linear models with a diverging number of covariates. *Biometrics*, 79(1), 344–357. <https://doi.org/10.1111/biom.v79.1>
- Xing X., Zhao Z., & Liu J. S. (2023). Controlling false discovery rate using gaussian mirrors. *Journal of the American Statistical Association*, 118(541), 222–241. <https://doi.org/10.1080/01621459.2021.1923510>
- Yan Z., Zachrisson K. S., Schwamm L. H., Estrada J. J., & Duan R. (2023). A privacy-preserving and computation-efficient federated algorithm for generalized linear mixed models to analyze correlated electronic health records data. *PloS One*, 18(1), e0280192. <https://doi.org/10.1371/journal.pone.0280192>

- Zang C., Hou Y., Schenck E., Xu Z., Zhang Y., Xu J., Bian J., Morozuk D., Khullar D., Nordvig A., Shenkman E., Rothman R., Block J., Lyman K., Zhang Y., Varma J., Weiner M., Carton T., Wang F., & Kaushal R. (2023). Risk factors and predictive modeling for post-acute sequelae of SARS-CoV-2 infection: Findings from EHR cohorts of the recover initiative. *Research Square*. <https://doi.org/10.21203/rs.3.rs-2592194/v1>
- Zhang H., Zang C., Xu Z., Zhang Y., Xu J., Bian J., Morozuk D., Khullar D., Zhang Y., Nordvig A. S., Schenck E. J., Shenkman E. A., Rothman R. L., Block J. P., Lyman K., Weiner M. G., Carton T. W., Wang F., & Kaushal R. (2023). Data-driven identification of post-acute SARS-CoV-2 infection subphenotypes. *Nature Medicine*, 29(1), 226–235. <https://doi.org/10.1038/s41591-022-02116-3>